



Toegepaste Micro-economie: Public economics

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Week 2:
Externalities

Lecture 2.5

Today: Applications of our theory / framework to

- Education (Slides added after lecture)
 - Government intervention in education: why?
→ applying the framework
 - Public provision vs public production
- ‘Tragedy of the commons’
 - Specific type of negative externalities:
Common-pool problem

Externalities: special case

What is common to these issues?:

- Dwindling populations of fish
- Traffic jams
- Strongly reduced flow in China's Yangtze river
- Global warming
- Overcrowded beaches, parks, shopping malls, etc.
- Slow wifi in public places

→ All (partially) caused by *common-pool problem*

Common-pool problem

Common-pool (or –resource) problem can arise when resources are Rival and Non-excluded

- Where ‘non-excluded’ is often caused by insufficiently well-defined and enforced property rights

Rival implies that the marginal cost of more producers / consumers is positive

Non-excluded implies that these cost are not taken into account by individual producers / consumers

Result: **Tragedy of the commons**

Common-pool problem

Tragedy of the commons:

Inefficiencies that arise because individuals do not take into account that their usage of a shared resource affects the costs and/or benefits of other users

Common-pool problem: example

Ork is a fishing community with N inhabitants who only care about money

Each day, each inhabitant of Ork can:

1. Undertake some paid work at wage w , or
2. Rent a boat at cost c , and go fishing.
 - The amount of fish that a fisher catches, F , decreases in the number of fishing-boats B on the lake
→ $F(B)$, where $F'(B) < 0$.
 - Assume: Ork has no market power on fish market
Market price is 1 per fish.

Consequences:

- Revenue from fishing is $F(B)$
- All efficiency effects on producer's side, no effects on consumer surplus

Common-pool problem: example

Market equilibrium number of boats B^P is given by

$$F(B^P) - c - w = 0$$

→ Income from fishing equals outside wage

Socially efficient number of boats B^S :

→ Maximise total income: $\text{Max}_B B[F(B) - c] + (N - B)w$

Which yields first-order condition for B^S :

$$F(B^S) - c - w + B^S F'(B^S) = 0$$

As $B^S F'(B^S) < 0$, we have that $B^P > B^S$

Common-pool problem: example

Why is $B^P > B^S$?

Because in deciding between working and fishing, individuals do not take into account that the presence of their boat on the lake reduce the quantity of fish caught by others.

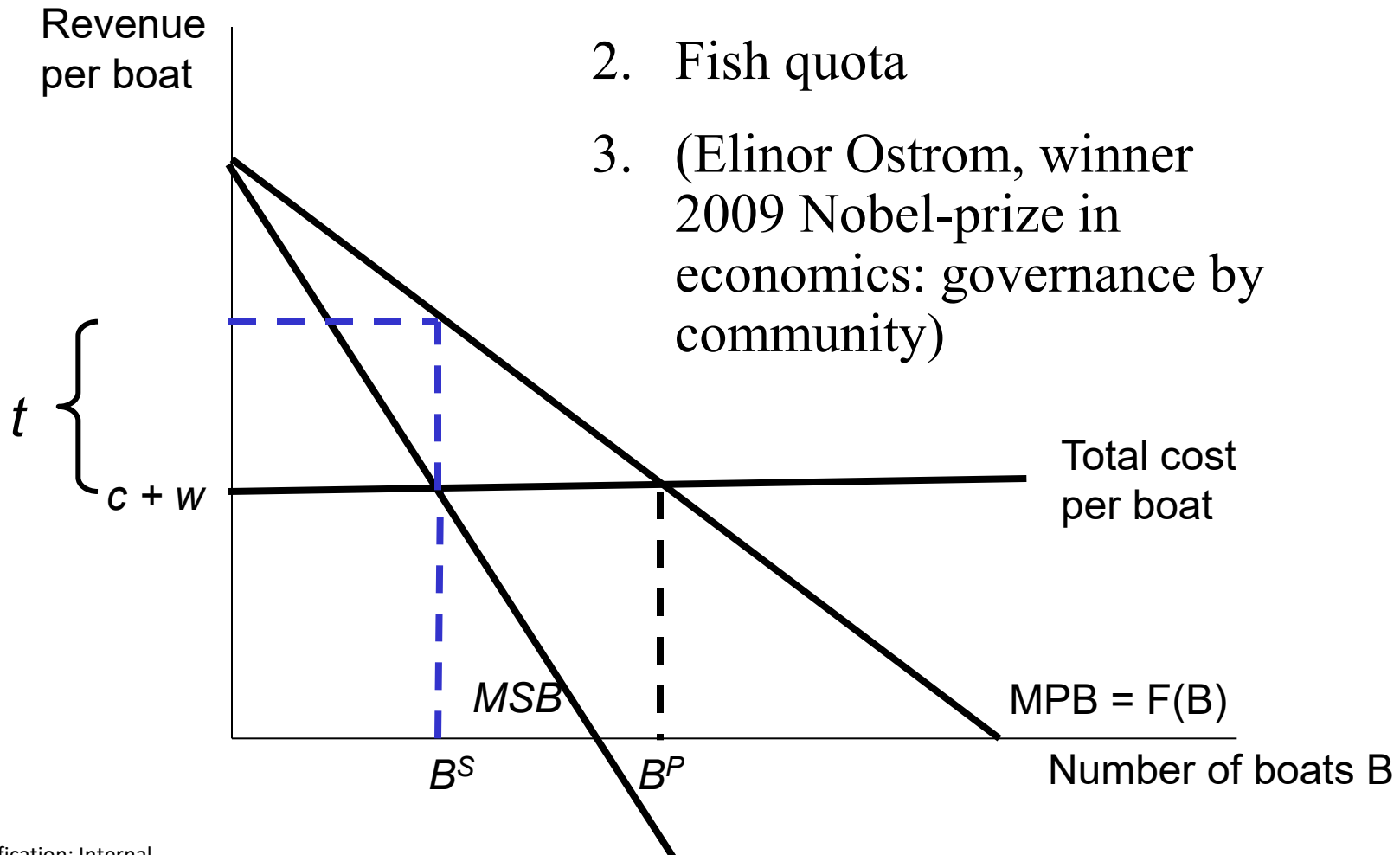
→ This is a negative externality!

Common-pool problem: example

In a figure:

How to get to B^S ?

1. Pigouvian tax per boat $t = -B^S F'(B^S)$
2. Fish quota
3. (Elinor Ostrom, winner 2009 Nobel-prize in economics: governance by community)



Common-pool problem

Tragedy of the commons arises when individuals do not take the effect of their actions on others into account.

Elinor Ostrom (winner 2009 Nobel-prize Economics) has shown that small communities often set up elaborate governance schemes to overcome the common pool problem, and can be successful
→ Coase Theorem (often with implicit contracts)

Concluding remarks

Many policy issues relate to externalities

→ Many examples of choices that affect others, outside the market mechanism

Moreover, some institutional / policy choices have created or enhanced externalities:

- Tax system and poverty alleviation

→ Your choice to invest in productivity-increasing education affects my tax rate

- Collective (health) insurance:

→ My decision not to sport or exercise may affect your insurance premium

Big issues, but also small → read Frank & Cartwright's chapter on externalities (ch 16 (18) in old (new) edition)

Looking ahead

Markets under asymmetric information

- Moral hazard
- Adverse selection
- Application to insurance

- Why millions of people in the US used to have no health insurance
- Why there is no privately provided poverty insurance
- Why the average time-to-graduation of students in NL is probably inefficiently low